

Quick Check

Lesson 3: Growth Curves: Why Do Vertebrates and Soft-Bodied Animals Grow Smoothly in an S-Shape While Insects with Hard Shells Grow in Steps?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.2: Growth and Development in Animals | Lesson 3

A Kenyan broiler chick has no rigid outer shell like a grasshopper's exoskeleton. What does this suggest about a graph of the chick's mass against time over 10 weeks?

- A** A smooth, continuous curve with no sudden jumps
- B** A staircase with flat parts and sudden jumps
- C** A straight line rising at the same rate every week
- D** A flat line because the chick only changes after moulting

A broiler chick from a Kiambu farm has a mass of 900 g at week 5 and 1500 g at week 7. Using $\text{Growth rate} = (\text{final value} - \text{initial value}) / \text{time interval}$, what is its growth rate between week 5 and week 7?

A 600 g/week

B 200 g/week

C 750 g/week

D 300 g/week

A grasshopper nymph's body length stays at 12 mm from day 10 to day 19. On day 20 it moults and its length jumps to 16 mm. Which explanation fits this stepped pattern?

- A** The nymph melts and rebuilds its whole body at each moult
- B** The hard exoskeleton cannot stretch, so body size increases only after the old one is shed
- C** The old skin peels off like sunburned skin, so the nymph is always growing underneath
- D** The nymph stops feeding between moults, so it cannot grow

A chick's growth rate is 240 g/week in weeks 3-6, 80 g/week in weeks 7-8 and about 3 g/week in weeks 9-10. Which is the best biological reason for growth slowing and then almost stopping?

- A** Cells divide faster and faster as the chick gets older
- B** The chick develops a hard exoskeleton that cannot stretch
- C** Resources and genetic limits restrict growth as the chick nears its adult size
- D** The chick sheds its skin, which removes mass

On the Driving Question Board, the question 'Do all animals grow the same way?' is to be moved to 'Answered'. Which evidence best supports moving it?

- A** Both a chick and a grasshopper get bigger as they get older
- B** A chick's mass graph is a smooth S-curve, while a grasshopper's length graph is a staircase with flat parts between moults
- C** A chick and a grasshopper both start life very small
- D** A chick's graph and a grasshopper's graph both have labelled axes

Which list gives the four phases of a sigmoid (S-shaped) growth curve in the correct order from birth to adult size?

- A** Lag, deceleration, exponential, stationary
- B** Stationary, lag, exponential, deceleration
- C** Lag, exponential, deceleration, stationary
- D** Exponential, lag, stationary, deceleration

A Friesian calf from Nyahururu rises from 40 kg to 70 kg in 10 weeks. What is its average growth rate?

A 30 kg/week

B 11 kg/week

C 0.3 kg/week

D 3 kg/week

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).